Verification status

Repository source: VERIFICATION_STATUS.md

Release: v0.1.0 (September 6, 2026)
Last updated: September 6, 2026
Named author: Bruce J. Swihart
AI system disclosed: OpenAI ChatGPT (GPT-5.6 Sol Pro), accessed August-September 2026
Canonical repository: https://github.com/swihart/minvol-degree3-spectral-bound
Versioned release: https://github.com/swihart/minvol-degree3-spectral-bound/releases/tag/v0.1.0

Current designation: AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification.

Candidate statement

The repository studies the proposed lower bound

$$\mathrm{Vol}(K) \geq \frac{\pi}{1914}\left(259 - 27\sqrt{\frac{15183}{17303}}\right) d^3 \approx 0.383602704709068\, d^3$$

for every three-dimensional convex body of constant width d.

This coefficient is larger than Nishioka's published coefficient $4\pi/33 \approx 0.380799109526$, but it is numerically weaker than the much newer public HYRA claims. The proposed contribution is the independent degree-3 spectral mechanism, not the strongest currently claimed coefficient.

Evidence in release v0.1.0

Layer	Evidence	Status
Published framework	Nishioka's support-function formulation, volume identity, spectral mode calculation, and smooth approximation	Sourced
Exact tensor algebra	General seven-parameter trace-free cubic checked by exact SymPy arithmetic	Passed internally
Independent exact formulation	A second determinant formulation checked symbolically on the sphere	Passed internally
Pairing enumeration	All contraction pairings for the six quartic moments counted exhaustively	Passed independently in Python and R
Numerical convention checks	Deterministic quadrature, random tensors, and rotational tests	Passed independently in Python and R
Human-readable proof	Bridge lemmas and the degree-3 tensor identity are written line by line	Drafted and internally audited
Integrated manuscript	Complete LaTeX research note compiles to PDF	Passed internally and approved by the named author

Layer	Evidence	Status
Hosted clean-environment checks	GitHub Actions runs Python, R, paper, Markdown-PDF, metadata, release, and PDF preflight checks	Required to pass on the exact release-candidate commit and version tag
Fresh-clone reproducibility	A separate clone of commit `96cea5727507` ran all checks and builds	Passed; rebuilt tracked tree remained byte-clean
Authorship and AI provenance	Full author name and the AI system, scope, access period, and limits are recorded	Documented
Citation and licensing	`CITATION.cff`, release URLs, release version/date, and separate prose/software license terms are present	Documented for v0.1.0
External mathematical review	Review by an independent subject-matter expert	Not yet completed
Peer review	Journal or conference peer review	Not yet completed

The detailed claim-by-claim record is in `proof/PROOF_LEDGER.md`. The dependency structure is in `proof/CLAIM_DEPENDENCIES.md`, the internal audit is in `proof/PROOF_AUDIT.md`, and the AI-use record is in `AI_ASSISTANCE.md`.

What a green GitHub Actions run means

The workflow in `.github/workflows/verification.yml` runs three jobs:

1. exact and numerical Python checks;
2. independent base-R checks; and
3. clean builds and preflight checks of the manuscript, metadata, release fields, and all Markdown-derived PDFs.

A green workflow establishes that the checked computations, metadata checks, and document builds run successfully in the recorded CI environments. It does **not** establish that the theorem has been independently proved, certified, or peer reviewed.

Recorded clean-clone result

Commit `96cea57275070f21babf30477ab1b128ec0f5eb6` was cloned into a new temporary directory on 2026-09-04. All Python and R checks ran, all tracked PDFs rebuilt, and the resulting tracked working tree was clean. The local Mac lacked Poppler, so basic PDF and checksum checks were performed locally and the full PDF preflight was supplied by the green GitHub Actions document job. See `CLEAN_CLONE_CHECK.md`.

The exact v0.1.0 release-candidate commit must pass one additional clean-clone run after all versioned metadata is committed and before the tag is created. No tracked file may change between that successful test and creation of the tag. After the tag is pushed, the tag-triggered three-job workflow must also pass before the GitHub release is published. The release page should record the tested commit and final clean-clone result.

Language approved for public use

Use:

> AI-assisted, non-peer-reviewed research draft seeking independent mathematical verification.

Do not use:

> Peer-reviewed theorem, certified proof, independently verified theorem, or accepted result.

Correction policy

The public version history should preserve every released draft. If an error is found, document it promptly, retain the earlier release, and issue a corrected version with a new tag. Material corrections should be summarized in the release notes and in the repository README.